Practical 4

* **Aim:-** Perform the following:- (a) Altering a Table, Dropping/Truncating/Renaming Tables, Backing up/Restoring a Database.

* **Theory:-** Database tables are not always fixed after creation. As requirements change, we may need to add new columns, remove unnecessary tables, rename tables, or delete all records. SQL provides commands to perform these operations efficiently. In addition, databases should be backed up regularly so that data can be restored if it is lost or corrupted.

(a) Altering Table:- The ALTER TABLE command is used to modify the structure of an existing table.
(i) Add new columns
(ii) Modify the datatype or size
(iii) Rename columns
(iv) Delete columns
(v) Add or remove constraints such as Primary Key or Foreign key.

Ex:- Create table STUDENTINFO
(SID INT, SNAME VARCHAR(20), BRANCH VARCHAR (20), CITY VARCHAR (20));
(i) Adding New column

ALTER table STUDENTINFO
ADD AGE VARCHAR(20);

(ii) Modifying the size
modify SNAME VARCHAR(100);

(iii) Changing column
change BRANCH Courseprogram VARCHAR
(20);

(iv) Dropping a column
Drop Age;

* Truncating a Table :- The Truncate table command deletes allrecords from a table but keeps the table structure.

Ex:-
truncate Student_info ;

* Dropping table:- Drop table command removes the entire table and all its data permanently.

Ex:-
drop table STUDENTINFO;

* Renaming a Table :- Rename table command changes the table name.

Ex:- rename table STUDENTINFO to
Student_Info;

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⑤ Backing up database :- A backup creates a copy of database so that it can be recovered later.

⑥ Restoring a Database :- Restoring recreates the database from the backup files.

* Conclusion :- In this practical, we learned how to modify table using Alter table, permanently remove tables using Drop table, delete records using truncate and rename table using Rename table. We learned to create backup and restore database.

Practical 5

Aim:- Implementation of different types of operators in SQL.

Theory:- SQL operators are special symbols or keywords used in SQL statements to perform operations on data stored in database tables. They are mainly used in the WHERE clause to filter records and in the SELECT clause to perform calculations.

① Logical Operators:- Logic operators are used to combine multiple conditions

(i) AND = Returns records only when all conditions are true.

Ex:- Select * from Employee
where Dept = 'MCA' AND Salary>=20000;

(ii) OR = Returns records where the one condition is atleast true.

Ex:- Select * from Employee
where Dept = 'MCA' OR Salary >20000;

(iii) Not = Returns records where the condition is false

Ex:- Select * from Employee
where NOT Dept = 'MCA';

② Special Operators:-

(i) Between :- Also known as range operator, Used to select values within a specified range.

Ex:- Select * from Employee
where Salary between 25000 AND 35000;

This query returns employees whose salary lies between 25000 and 35000.

(ii) Like :- Also known as pattern matching operator Used to search for a specific pattern in text values.

Ex:- Select * from Employee
where Emp_Name LIKE 'R%D%';

This query finds employee names that starts with R and contain D ~~in~~ later in the name.

③ Comparision Operators:-

Used to compare values:-

(i) < less than

Ex:- Select * from Employee where salary<30000;

(ii) > greater than

Ex:- Select * from Employee where salary>30000;

(iii) = Equal to

Ex:- Select * from Employee where salary=30000;

④ Arithmetic Operators :- Used to perform mathematical calculations on numeric columns.

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(i) Addition (+)

Ex:- Select Emp-Name, Salary+5000 As Increased-Salary from Employee;

(ii) Subtraction (-)

Ex:- Select Emp-Name, Salary, Salary - 1000 As Decreased - Salary from Employee;

(iii) Multiplication (*)

Ex:- Select Emp-Name, Salary, Salary *2 As Doubled_Salary from Employee

(iv) Division (/)

Ex:- Select Emp-Name, Salary, Salary /2 As Half-Salary from Employee

These queries display calculated salary values without modifying the original data.

* Conclusion :- Thus, different SQL operators such as logical, comparison, special operators and arithmetic operators were successfully implemented on the Employee table. These operators helped in filtering records and performing calculations on data efficiently.